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Assignment – 7

Aim

To implement and analyze probability distributions—specifically Binomial,

Poisson,
and Normal distributions—using Python, and to understand their characteristics, differences, and real-world applications.

1.**Difference between Binomial, Poisson, and Normal Distributions**

Feature	Binomial Distribution	Poisson Distribution	Normal Distribution
Type	Discrete	Discrete	Continuous
Use Case	Fixed number of trials	Rare events over time/space	Natural continuous data
Parameters	n (trials), p (probability)	λ (average rate)	μ (mean), σ (std deviation)
Example	Tossing a coin	Number of calls per hour	Height of people
Shape	Can be symmetric/skewed	Right-skewed	Symmetric (bell-shaped)

2. **Real-world significance of Poisson distribution**

Examples:

Number of calls in a call center per minute Number of accidents on a road per day Number of typing errors per page ✓It helps in: Predicting probabilities of rare events Planning resources (like staffing in hospitals or call centers) Risk analysis in industries

3. **Central Limit Theorem (CLT) and Normal Distribution**

The Central Limit Theorem (CLT) states: X ¯~N(μ, n

σ) ✓Meaning: When you take large samples from any distribution, The sample mean will be approximately normally distributed, Even if the original data is not normal.

Relation to Normal Distribution✓CLT explains why Normal distribution appears everywhere It allows us to use Normal distribution for: Estimation Hypothesis testing

4. **Why Normal distribution is widely used**

The Normal distribution is popular because: It naturally occurs in many real-world datasets (height, weight, marks, errors, etc.) It is mathematically simple and well-understood Many statistical methods assume normality: Regression Hypothesis testing Machine learning algorithms Due to CLT, it works well even when original data is not normal

5. **Skewness in distributions**

Skewness means asymmetry in a distribution. Types: Positive skew (right-skewed) → tail on right side

Negative skew (left-skewed) → tail on | Distribution | Skewness | | ---------- | ---------- | | Binomial | Can be skewed (depends on p) | | Poisson | Positively skewed | | Normal | No skew (perfectly symmetric) | left side Zero skew → symmetric

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm


n = 10
p = 0.5

x_binom = np.arange(0, n+1)
y_binom = binom.pmf(x_binom, n, p)

plt.figure()
plt.bar(x_binom, y_binom)
plt.title("Binomial Distribution")
plt.xlabel("Number of Successes")
plt.ylabel("Probability")
plt.show()


# ----------- Poisson Distribution -----------
lam = 3   # average rate

x_pois = np.arange(0, 15)
y_pois = poisson.pmf(x_pois, lam)

plt.figure()
plt.bar(x_pois, y_pois)
plt.title("Poisson Distribution")
plt.xlabel("Number of Events")
plt.ylabel("Probability")
plt.show()


# ----------- Normal Distribution -----------
mu = 0       # mean
sigma = 1    # standard deviation

x_norm = np.linspace(-4, 4, 100)
y_norm = norm.pdf(x_norm, mu, sigma)

plt.figure()
plt.plot(x_norm, y_norm)
plt.title("Normal Distribution")
plt.xlabel("Values")
```

```
plt.ylabel("Probability Density")
plt.show()
```

Conclusion: